

RESEARCH ARTICLE

Effect of Dietary Onion Peel Powder (*Allium cepa* L.) as a Feed Additive on Production Performance in Broiler

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ABSTRACT

This study aims to evaluate the use of onion peel powder (*Allium cepa* L.) as a feed additive on feed intake, weight gain, Feed Conversion Ratio (FCR), Production Index, and Income Over Feed Cost (IOFC) in broilers. A total of 144 Lohmann strain broiler (unsexed) were treated from 1 to 35 days of age. There were 6 treatments with 4 replicates, consisting of 24 experimental cages, each filled 6 broilers. The experimental design used completely randomized design (CRD). The treatment groups were as follows: basal feed without antibiotics (P0-), basal feed + 0.1% antibiotics (zinc bacitracin) (P0+), basal feed + 0.25% onion peel (P1), basal feed + 0.50% onion peel (P2), basal feed + 0.75% onion peel (P3), basal feed + 1% onion peel (P4). The data were analyzed using ANOVA using Duncan's Multiple Range Test (DMRT). The results showed that the addition of onion peel powder has no significant effect on broiler production performance ($P>0.05$). The use of onion peel powder at dose of up to 1% did not cause negative effects on broilers. The application of onion peel powder at a dose of 0.50% (P2) showed an appropriate dosage for use as a feed additive in broiler feed.

KEYWORDS

Additive; Broiler; Onion peel; Production performance

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1. Introduction

Indonesia is an agrarian country rich in natural resources, one of which is the livestock sector. The development of broiler in Indonesia has seen an increase in production and consumer demand every year. Factors that influence the success of livestock businesses include seeds, feed, maintenance management, and biosecurity. Feed plays a very important role in the broiler maintenance process. By paying attention to the right nutritional content, optimal productivity can be achieved. Efforts made by farmers to overcome high feed costs include the addition of growth promoters or synthetic antibiotics. The impact of antibiotic use can be harmful to consumers, prompting the government to issue Minister of Agriculture Regulation No. 14/PERMENTAN/PK.350/5/2017 on the classification of veterinary drugs, which prohibits the use of antibiotic feed additives. The efforts of farmers to reduce the use of antibiotics as feed additives involve the addition of phytobiotics. Phytobiotics are plant components that can be used as feed additives that come from vegetables, fruits, medicinal plants, and grains (Kari *et al.*, 2022). Onion peel is one phytobiotic that can be used as an alternative. According to the Ministry of Agriculture's General Secretariat (2025), onion production in 2025 is projected to reach 32.33 tons. Fresh onions generate 16 g/kg of wet onion peel waste, 14 g/kg of dry onion peel, and 13.72 g/kg of onion peel flour (Jannah *et al.*, 2024). Onion peel (*Allium cepa* L.) has active compounds that help protect bulbs from physical and chemical damage. Flavonoids in onions determine the different compositions of peel and bulb colors; the yellow color in onions de-

termines the amount of quercetin and its derivatives, while the red color is caused by anthocyanins. Onions are the best source of flavonoids, including flavonols and anthocyanins (Celano *et al.*, 2021). Dried onion peel has a high percentage of aglycone flavonols, while the anthocyanin content is limited. Onion peel has the potential to be a good source of polarized waste. Adding onion peel to broiler can increase weight gain (Malematja *et al.*, 2022). Research conducted by Rahman *et al.* (2022) states that adding onion peel to broiler feed can improve meat quality. This study on the use of onion peel powder (*Allium cepa* L.) as a phytobiotic feed additive in broiler, specifically to comprehensively evaluate its effect on feed intake, weight gain, feed conversion ratio, production index, and Income Over Feed Cost (IOFC) under antibiotic-free conditions as a control feed.

This study is based on the necessity of safe, effective, and sustainable feed additives that are capable of replacing synthetic antibiotics in the poultry industry. The application of onion peel waste as feed powder is strategically valuable because it contains bioactive compounds such as flavonoids, saponins, and tannins that are potential antioxidants. The use of agricultural waste as feed ingredients also supports the efficiency and sustainability in livestock production systems. The aim of this study is to examine the potential of onion peel powder as a natural feed additive that can improve performance of broiler production. This study supports the development of more safe, efficient, and sustainable livestock systems.

2. Materials and Methods

2.1. Materials

The material used in this study was onion peel that had been processed into powder to be used as a feed additive mixed manually with commercial feed. The onion belongs to the Liliaceae family and is available at Malang City Market. Zinc bacitracin was used as an antibiotic administered in the control treatment. The feed that had undergone the mixing process was then tested in vivo on broiler. A total of 144 Lohmann strain broiler, which were not sexed, were raised for 35 days. Each treatment compartment was filled with 6 broiler selected at random. Broiler were given treatment from day 1 to day 35 during rearing. The cages used in this study were cages with rice husk litter, consisting of 24 experimental cages measuring 1x1x1 m per unit. Each pen was equipped with a feeding trough and drinking trough. Feed and drinking water were provided ad libitum and vitamins were administered. Vitamins (vitachick) were added to the drinking water to reduce stress levels in the broiler. Disinfectants were used to sterilize the cages and other equipment. The equipment used included digital scales, feeders, drinkers, and writing instruments.

2.2. Methods

Onion peel flour is produced using onion peels obtained from the market, which are then sorted to remove rotten and dirty peels; the clean onion peels are set aside and air-dried at room temperature. A total of 26 kg of fresh material is dried in an oven for 3 days, or 72 h, at 50°C until the moisture content drops below 10%. After drying, the moisture content is checked to determine the shrinkage using a moisture balance, followed by grinding until a powdery texture is achieved. The onion peel powder is then sieved using a 230-mesh sieve, and the sieved product is used as a feed additive.

This research used a field experimental method with treatment of broiler from 1-35 days of age, designed using a completely randomized design (CRD) with 6 treatments and 4 replicates, resulting in 24 experimental cages. Production performance data based on the average per bird. If significant differences were found, analysis was performed using Duncan's Multiple Range Test (MDRT) using Microsoft Excel. Each experimental cage unit contained 6, requiring a total of 144 broiler (unsexed). This study was approved by the University Ethics Committee (approval No. 42-KEP-FKHUB-2025). The treatments given to the broiler were as follows (Jannah, Sudjarwo, and Sjojfan, 2024):

P0 (-) : Basal feed without antibiotics

P0 (+) : Basal feed + antibiotics (zinc bacitracin) 0.1%

P1: Basal feed + 0.25% onion peel powder

P2: Basal feed + 0.50% onion peel powder

P3: Basal feed + 0.75% onion peel powder

P4: Basal feed + 1% onion peel powder

The nutritional content of basal feed can be seen in Table 1.

Table 1. Nutritional Content of Basal Feed

Nutritional Content	BR 1
Net protein (%)	21,00-23,00
Moisture (%)	Max. 12,00
Fat (%)	Min. 5,00
Crude fiber (%)	Max. 5,00
Ash (%)	Max. 7,00
Calcium (%)	0,8-1,1
Phosphor (%)	Min. 0,50
EM (Kkal/kg)	Min. 3000
Aflatoxin (g/Kg)	Max. 50

Source: PT. Japfa Comfeed Indonesia, Tbk.

Research data was collected by weighing DOCs to obtain the coefficient of variation. The amount of feed given and leftover feed was weighed daily to calculate feed consumption. Body weight was measured weekly to calculate weight gain, which could later be used as comparative data in the Feed Conversion Ratio (FCR), to determine the Production Index (PI) and Income Over Feed Cost (IOFC). The feed consumption formula according to Gultom, Sjojfan, and Sudjarwo (2022) is as follows:

Feed consumption=amount of feed given (g) -amount of feed leftovers (g)

The body weight gain formula according to Gultom, Sjojfan, and Sudjarwo (2022) is as follows:

Weight gain=final weight (g) -initial weight (g)

The feed conversion ratio according to Gultom, Sjojfan, and Sudjarwo (2022) is as follows:

$$\text{Feed conversion ratio} = \frac{\text{feed intake (g)}}{\text{weight gain (g)}}$$

The production index (PI) formula according to Jabbar and Syam (2024) is as follows:

$$\text{Production index} = \frac{(100\% - \text{Dep}\%) \times \text{BW}}{\text{FCR} \times \text{A/U}} \times 100\%$$

Notes:

Dep% = Depletion percentage

BW = Body weight

FCR = Feed conversion ratio

A/U = Harvest age

The Income Over Feed Cost (IOFC) formula according to Santani et al. (2023) is as follows:

IOFC=sales-feed costs

2.3. Statistical Analysis

This study used a field experimental method with treatments on broiler from 1 day to 35 days of age using a Completely Randomized Design (CRD) with 6 treatments and 4 replicates. The data obtained were tested using Analysis of Variance (ANOVA). The production performance variables are based on the average per bird. In the statistical analysis of this treatment, the formula (Nugraha dkk, 2024) was used.

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

Notes:

Y_{ij} : Observation value from treatment i and repetition j

i: Indicates the number of t treatments

j : Indicates the number of r repetitions

μ : General mean value

τ_i : Effect of treatment i

ϵ_{ij} : Experimental error

3. Results and Discussion

3.1. Effect of Dietary Onion Peel Powder (*Allium cepa* L.) as a Feed Additive on Feed Consumption in Broiler

Feed consumption is important for determining feed palatability in relation to broiler growth performance. Feed consumption is obtained by weighing the amount of feed given and then subtracting the amount of feed remaining. The results of the study on the effect of adding onion peel on broiler feed intake can be seen in Table 2. The average feed intake values for broiler

from highest to lowest are as follows: P0(+) (3004.375±207.06), P0(-) (2887.7±102.28), P1 (2858.25±92.39), P2 (2836.5±198.1), P4 (2783.08±277.42), P3 (2673.58±106.04). Statistical analysis showed that the addition of onion peel powder as a feed additive

had no significant effect ($P>0.05$) on the feed intake of broiler. This is because the feed and energy content used are the same, so the resulting feed intake tends to be roughly the same (Barzegar *et al.*, 2020).

Table 2. Effect of Onion Peel Powder on Production Performance in Broilers

Variable	Treatment (Data Mean±SD)						P value
	P0 (-)	P0 (+)	P1	P2	P3	P4	
Feed Intake (g/bird)	2887.7±102.28	3004.375±207.06	2858.25±92.39	2836.5±198.1	2673.58±106.04	2783.08±277.42	>0.05
Body Weight Gain (g/bird)	1653.62±146.71	1684.95±75.51	1524.37±14.03	1625.29±110.79	1425.04±154.3	1530.04±150.9	>0.05
Feed Conversion Ration (FCR)	1.70±0.14	1.73±0.08	1.82±0.06	1.69±0.07	1.85±0.13	1.76±0.04	>0.05
Production Index	287.5±44.06	284.5±16.54	247.8±15.17	282.8±23.14	224.8±33.76	251±20.14	>0.05
Income Over Feed Cost (IOFC) (Rp/bird)	11506.77±237.11	12295.93±2118.16	12364.12±1337.13	11718.25±2142.91	12110.95±386.97	11373.2±1957.13	>0.05

Notes: P0(-): Basal feed without antibiotics, P0(+): Basal feed + 0.1% antibiotic (zinc bacitracin), P1: Basal feed + 0.25% onion peel, P2: Basal feed + 0.50% onion peel, P3: Basal feed + 0.75% onion peel, P4: Basal feed + 1% onion peel

The results of the study stated that P0(+), the treatment of adding 0.1% antibiotics to the feed, had the highest average, but in the feed treatment, the results showed that P1, the treatment of adding 0.25% onion powder to the feed, had the highest average. This is because onion peel powder contains bioactive compounds including flavonoids, saponins, and tannins that can affect palatability, digestive tract health, and feed flavor. Flavonoids can modulate the microbiota to improve the intestines and nutrient digestion (Prihambodo *et al.*, 2020). Saponins have the potential to increase the secretion of intestinal microbiota enzymes and can cause a surfactant reaction that can change the texture of food (Obianwuna *et al.*, 2024). Tannins have binding properties with proteins and minerals as well as an astringent taste that can reduce feed palatability and decrease feed consumption when used in high doses (Galamatis *et al.*, 2025). The complete components of flavonoids, saponins, and tannins can improve the digestive system and health of livestock, but at normal doses, they do not cause significant differences in livestock.

3.2. Effect of Dietary Onion Peel Powder (*Allium cepa* L.) as a Feed Additive on Weight Gain in Broiler

Weight gain is an effective indicator for assessing growth rate in broiler farming. Weight gain is calculated by subtracting the initial weight from the final weight. The results of the study on the effect of adding onion peel on the weight gain of broiler can be seen in Table 2. The average weight gain of broiler from highest to lowest in sequence is P0 (+) (1684.95±75.51), P0 (-) (1653.62±146.71), P2 (1625.29±110.79), P4 (1530.04±150.9), P1 (1524.37±14.03), P3 (1425.04±154.3). Statistical analysis showed that the addition of onion peel powder as a feed additive had no significant effect ($P>0.05$) on the weight gain of broiler. This is because there was no significant difference in feed intake. Flavonoids in onion peel can improve digestive health by improving intestinal morphology and enhancing nutrient absorption. Flavonoids also act as antioxidants that neutralize free radicals, thereby reducing oxidative damage. Feed energy in livestock consuming flavonoids can be utilized more efficiently for protein and tissue synthesis, which can increase body weight gain (Abdel-Latif *et al.*, 2021). In this study, there was no difference, which may be due to the administration of the same feed nutrients and the use of feed additives at low doses.

The results of the study stated that P0 (+) treatment with 0.1% antibiotic feed addition had the highest average, but in the feed treatment, P2 treatment with 0.5% onion powder feed addition

had the highest average. Saponins in onion peel can stimulate the secretion of digestive enzymes and gut microbiota, thereby increasing nutrient utilization and growth. The saponins consumed by livestock depend on the dosage and composition of other feed additive compounds (Ren *et al.*, 2025). This occurs because the flavonoids, saponins, and tannins in onion peel can improve nutrient supply, intestinal health, and energy metabolism in broiler. A good balance of gut microbiota can increase feed utilization efficiency so that more nutrients are converted into body weight (Prihambodo *et al.*, 2020).

3.3. Effect of Dietary Onion Peel Powder (*Allium cepa* L.) as a Feed Additive on Feed Conversion Ratio in Broilers

Feed Conversion Ratio (FCR) plays an important role because the lower the FCR value, the better the feed conversion and the lower the feed costs during maintenance (Nuryati, 2019). The results of the study on the effect of adding onion peel on the hemoglobin levels of broiler can be seen in Table 2. The average FCR values of broiler from highest to lowest were P3 (1.85±0.13), P1 (1.82±0.06), P4 (1.76±0.04), P0 (+) (1.73±0.08), P0 (-) (1.70±0.14), P2 (1.69±0.07). Statistical analysis showed that the addition of onion peel powder as a feed additive had no significant effect ($P>0.05$) on the FCR of broiler. This is because onion peel contains flavonoids that improve intestinal morphology, thereby increasing the area available for nutrient absorption, such as amino acids and glucose. Improved absorption enhances digestive efficiency and nutrient utilization, allowing energy to be redirected toward muscle protein synthesis and livestock tissue growth (Prihambodo *et al.*, 2020). This study indicates that there are no significant differences, this is because feed intake and body weight gain also showed no differences, as the nutrients and feed used were the same.

The results of the study stated that treatment P3, which added 0.75% onion powder to the feed, had the highest average but was still within normal limits. This is because high doses of tannins can form complexes with proteins and digestive enzymes in the intestine, thereby inhibiting protein digestion and amino acid absorption (Pertwi *et al.*, 2023). Flavonoids, saponins, and tannins in onion peel can increase intestinal metabolism and nutrient breakdown, while tannins in certain doses can interfere with nutrient absorption (Abdelli, Solà-Oriol, and Pérez (2021). This can occur because, according to research conducted by Zhang *et al.* (2016), saponins increase weight gain through improved digestion and immunomodulation. Saponins are important for undergoing a growth phase with high nutrient requirements and